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MTMS: Age-Aware Multi-Task Multi-Layer Stacking for SSD Failure Prediction

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***Abstract*—Heterogeneous data modes, varying SSD ages, and hidden data distribution shifts pose challenges for accurate SSD failure prediction. To address these challenges, we propose MTMS, a novel multi-task, multi-layer stacking framework designed for proactive SSD failure prediction. MTMS leverages age-aware mechanism and multi-task learning to divide the prediction task into subtasks based on SSD usage age, training specialized sub-models within each group. These sub-models are stacked to form an intermediate layer, which is then combined with additional classifiers in a second stacking process. A voting mechanism aggregates predictions from all classifiers, ensuring more robust and accurate results. Our experimental results on the Alibaba SMART dataset demonstrate that MTMS significantly outperforms traditional machine learning algorithms and three state-of-the-art methods, delivering more precise predictions and reducing maintenance costs.**

Keywords—Solid State Drive, Failure Prediction, Feature Extraction, Machine Learning

I. Introduction

Approximately 80% of storage system failures are caused by hardware failures, with drive failure being the leading factor [1]. Solid-state drives (SSDs) have gradually become a viable alternative to mechanical hard disk drives (HDDs) due to their ultra-high performance, and are widely used in modern data centers. Therefore, ensuring the reliability of SSDs has become an important issue.

As a relatively new type of storage device, NAND flashbased SSDs have received less attention regarding reliability compared to HDDs [2] – [5]. Existing studies either analyze SSD failure causes in controlled experimental environments or focus solely on a few fixed SMART attributes for SSD data analysis [6], [7]. However, real-world cloud storage systems must handle diverse workloads, unpredictable user operations, and numerous concurrent read/write requests, which limits the applicability of laboratory findings. Furthermore, features whose relevance to failure events varies over time (i.e., feature importance drift) are closely tied to changes in data distribution over long-term sequences. These factors collectively affect the portability and accuracy of prediction models, posing significant challenges for SSD failure prediction.

To address these challenges, this paper proposes a novel prediction framework, MTMS. MTMS introduces an ageaware feature extraction strategy, implemented via multi-task learning and enhanced through stacking techniques. The main contributions of this paper are as follows:

- We identify challenges in SSD failure prediction across varying types and ages, including data distribution changes and feature importance drift. We propose an ageaware feature extraction strategy that groups and extracts features of SSDs based on their age to mitigate noise interference.
- We introduce MTMS architecture that combines ageaware feature extraction with multi-task learning and multi-layer stacking. Extensive experiments on public datasets demonstrate MTMS ' s significant improvement in prediction accuracy for diverse SSDs.

The remainder of the paper is organized as follows. Section II summarizes related works, the proposed method is presented in Section III, Section IV provides the experimental setup and discusses the results, and finally, the work is concluded in Section V.

II. ReLated Work

Due to the multi-year span of SSD SMART sequences, data distribution changes over time, complicating the identification of failure-related features. Additionally, structural limitations of certain models hinder their ability to effectively extract features from such long-term sequences, leading to overfitting and poor generalization. To address these challenges, researchers have proposed various methods focusing on mitigating data imbalance, improving feature engineering, and optimizing models.

Liu et al. [8] introduced the MTHC method, which employs machine-level predictions to mitigate data imbalance. Zhou et al. [9] applied active learning techniques to monitor and augment positive samples. Zhou et al. [10] proposed a variational autoencoder (VAE)-based model trained solely on healthy SSD data to address the data imbalance problem.

In the feature engineering field, Wang et al. [11] highlighted the importance of optimal feature selection for enhancing prediction accuracy. Zhang et al. [12] proposed MVTRF, a method combining multi-input and group decision trees to capture long- and short-term data patterns. Xu et al. [13] introduced WEFR, which employs wear-leveling for ensemble feature ranking, automating SMART attribute selection to enhance resilience. Lu et al. [14] utilized performance metrics, positional data, and SMART features to form feature groups, improving feature selection efficiency via the J-Index method.

Ensemble learning techniques have also received considerable attention for their ability to improve model performance [15], [16]. However, many proposed approaches overlook the dynamic nature of SSD data, where the relevance of failure-related features can change over time due to factors such as SSD usage age and evolving data distributions [17]. This research gap motivated us to explore the age-aware mechanism, which enhances the robustness of extracted key features by capturing temporal variability, thereby mitigating the overfitting problem.

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III. Methodology

As shown in Fig. 1, the proposed MTMS framework integrates two core components: multi-task learning (MTL) module and multi-layer stacking (MLS) module.

Multi-task learning methods use parameter sharing and feature sharing among subtasks to learn generalized representations, thereby maximizing data utilization and enhancing generalization when data is scarce. This study applies a multitask learning method combined with an age-aware mechanism to address the heterogeneity of age in the SSD dataset (i.e., the MTL module). The age-aware mechanism classifies the data into different age groups according to usage duration. Each age group is treated as an independent sub-task, where feature extraction, re-weighting, and sub-model training are performed.

The MLS module enhances the integration of sub-models by employing a hierarchical stacking method. This approach generates a meta-classifier that combines the predictions of individual sub-models and integrates them with other classifiers that utilize diverse model architectures, including Random Forest, Gradient Boosted Decision Trees, and Neural Networks, each designed to capture complementary aspects of the input data. The final prediction result is produced through a voting mechanism, where the meta-classifier and other stacked classifiers collectively contribute to the decisionmaking process.

During the prediction phase, the input data skips additional grouping and is directly processed by the meta-classifier generated through the stacking method. This meta-classifier dynamically selects the appropriate sub-model to generate the final classification output using pre-trained knowledge.

A. Age-aware Mechanism

The age-aware mechanism is a core component of the proposed framework. Fig. 2 depicts the specific workflow of the age-aware mechanism within the MTL module, which segments input data into distinct age groups, adjusts feature importance, and refines group boundaries to ensure robust feature selection. The mechanism includes three main stages: initial age group segmentation, deduplication to merge similar groups, and optimized feature weighting for model training.

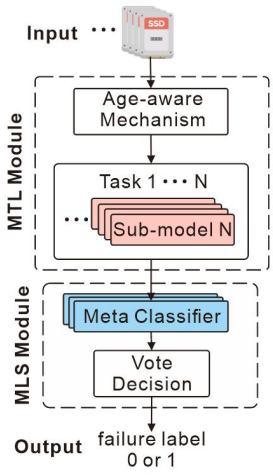


Fig. 1. Overall Architecture.

The first step is determining the initial segmentation intervals for age groups. Data is divided based on SSD types and Power-On Hours (POH) into intervals of 60, 90, 180, and 360 days. Kernel Density Estimation (KDE) is employed to analyze the data distribution, while change point detection identifies significant shifts in data behavior. Among the intervals tested, the 90-day interval achieves the best balance between interpretability and prediction performance, with the highest recall, accuracy, and F1 score, and stable False Alarm Rates (FAR), as shown in Fig. 3.

After initial segmentation, a deduplication mechanism refines the age groups to improve representation. Adjacent groups are compared using the Jaccard coefficient, which measures the similarity between their key feature lists. A predefined threshold guides the merging of similar groups, optimized through:

- **Density Analysis:** Calculating the mean and standard deviation of Jaccard similarities to determine a recommended threshold range (0.57 to 0.71).
- **Binary Search:** Iteratively refining the threshold within this range to maximize the model's F1 score.

This process ensures that the final age groups are distinct, representative, and free of redundancy. The optimization process is illustrated in Fig. 4.

Once the final age groups are defined, feature weights (x1, x2, . . . , xn) are reallocated based on their importance rankings. A linear transformation emphasizes the most critical features, maximizing the use of key information. These reweighted features are input into sub-models trained using Random Forest as the base model. The trained sub-models are stored in a dictionary indexed by SSD type and age group boundaries.

The integration of the age-aware mechanism within the MTL module facilitates the sharing of knowledge across age groups and enables parameter and feature sharing among age groups.

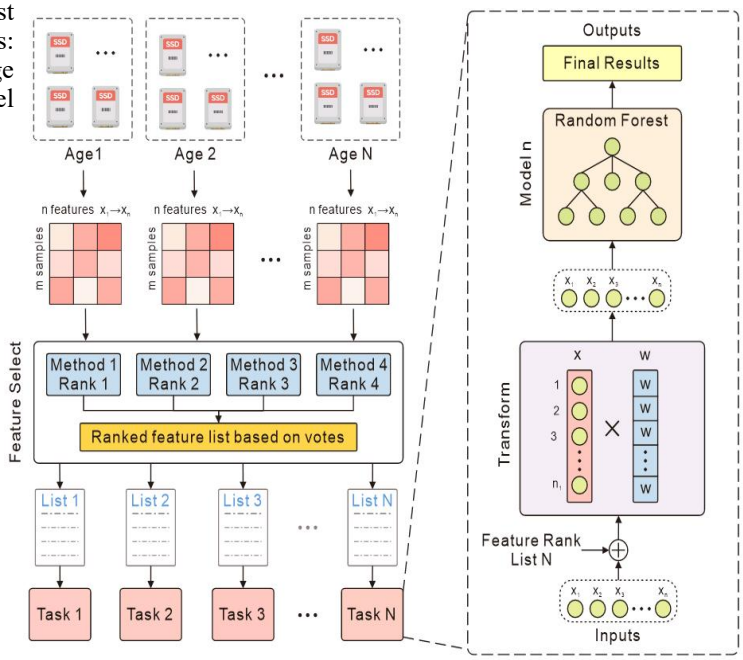
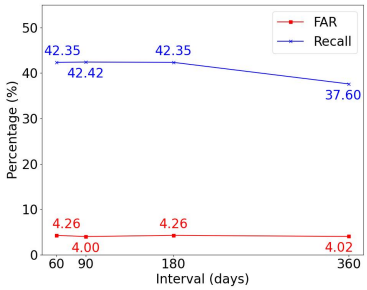
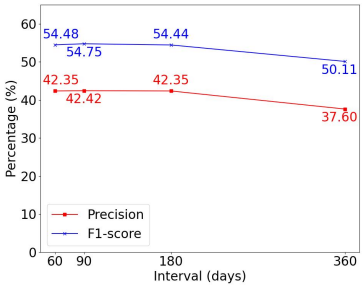


Fig. 2. Architecture of the MTL Module.

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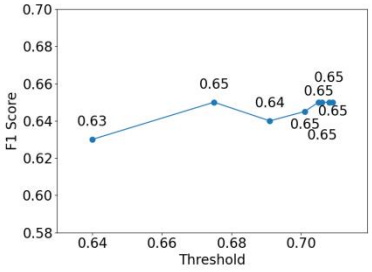


(a) Comparison of FAR and recall.

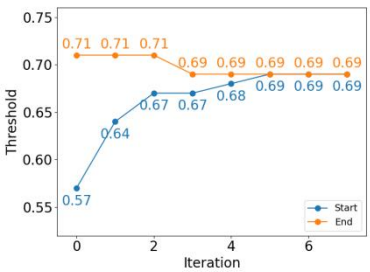


(b) Comparison of precision and F1-score.

Fig. 3. Comparison of prediction performance at different time segmentation intervals.



(c) F1-score of the model under different Jaccard thresholds.



(d) Changes in threshold intervals during the binary search process.

Fig. 4. Jaccard optimal threshold determination process.

IV. Experiments

A. Datasets

The dataset for this study was collected from Alibaba's industrial environment across five data centers between January 1, 2018, and December 31, 2019, and includes nearly 500,000 SSDs of six types from three manufacturers. It comprises SMART logs with 102 attributes and failure logs documenting 16,305 SSD failures, recorded every 24 hours. Detailed experimental data are provided in TABLE I. The homogeneous dataset is composed exclusively of SSDs of type MC1. The heterogeneous dataset encompasses all SSD types.

TABLE I. Experimental Dataset.

Dataset		Neg.	Pos.	Start	End
Homogeneous	***Train Set***	189152	1819	2018.7	2019.2
	Validation Set	189152	3312	2019.3	2019.7
	Predict set	189152	4725	2019.8	2019.12
Heterogeneous	***Train Set***	458751	4835	2018.7	2019.2
	Validation Set	458751	4219	2019.3	2019.7
	Predict set	458751	5508	2019.8	2019.12

B. Experimental Results and Analysis

1) Effectiveness of Age-aware Mechanism

We compared our proposed age-aware mechanism with five other feature-processed methods, including traditional approaches such as Spearman, Pearson, and Random Forest, alongside more recent techniques like J-Index [14] and WEFR [13]. The experimental design in this study was conducted on the heterogeneous datasets listed in Table 1, utilizing a Random Forest classifier with n estimators=1000 as the base model.

The results in TABLE II demonstrate the effectiveness of the age-aware mechanism in dynamically selecting features based on SSD age, achieving the highest F1-score of 47.8% and the lowest FAR of 0.3%. While traditional methods like Spearman and Pearson achieve high precision with increased FAR, and novel methods like J-Index and WEFR improve recall at the expense of precision, the age-aware approach strikes an optimal balance, reducing overfitting and enhancing prediction generalization.

TABLE II. Comparison of Age-aware Mechanism with Traditional and Recent Methods.

Methods	FAR(%)	Recall(%)	Precision (%)	F1-score(%)
Spearman	0.6	24.6	89.0	38.5
Pearson	1.1	22.0	78.4	34.4
RandomForest	1.6	30.5	78.2	43.9
J-Index	1.4	22.4	74.8	34.5
WEFR	1.5	26.8	77.8	39.8
Age-aware	**0.3**	**32.0**	**94.2**	**47.8**

2) Performance of MTMS Architecture

As shown in TABLE III, we compared the MTMS architecture against established methods, including SVM, GBDT, CNNLSTM [14], 1-class AutoEncoder(AE) [18], and MTHC [8], to evaluate its performance on both homogeneous and heterogeneous datasets. This evaluation aimed to highlight the benefits of the MTMS framework, particularly when enhanced by the age-aware feature extraction strategy.

Results on Homogeneous Datasets: On homogeneous datasets, the MTMS framework achieved superior performance compared to traditional and novel methods, with a FAR of 0.03% and a precision of 72.00%. Incorporating the age-aware mechanism further improved performance, with the complete MTMS framework achieving a precision of 76.67% and an F1-score of 30.87%, demonstrating its effectiveness in balancing recall and precision for SSD failure prediction.

Results on Heterogeneous Datasets: On heterogeneous datasets, the MTMS framework outperformed all compared methods, achieving a precision of 87.2% and a FAR of 2.0%. The age-aware mechanism enabled dynamic adaptation to diverse data distributions, ensuring accurate predictions across varying failure modes. These results highlight the MTMS framework's robustness and adaptability to complex, heterogeneous data environments.

V. Conclusion

In this paper, we introduce an age-aware mechanism, integrated into a multi-task multi-layer stacking framework, formed MTMS. The age-aware mechanism enhances the robustness of the SSD failure prediction scheme and reduces noise interference through age-based multi-task processing. The MTMS architecture, combined with this mechanism, mitigates the overfitting problems and adapts to diverse SSD patterns in cloud storage systems. In future work, we will

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TABLE III. COMPARISON OF METHODS' PERFORMANCE ON HOMOGENEOUS AND HETEROGENEOUS DATASETS.

Methods		**FAR(%)**		**Recall(%)**		**Precision(%)**		**F1-score(%)**	
Feature Engineering	***Model***	***Homo.***	***Hetero.***	***Homo.***	***Hetero.***	***Homo.***	***Hetero.***	***Homo.***	***Hetero.***
Feature Voting Ranking	SVM	1.44	11.2	20.00	40.5	5.78	54.0	8.97	50.6
Feature Voting Ranking	GBDT	0.20	2.1	20.00	27.9	30.30	80.7	24.10	41.5
Feature Extraction Based on J-Index	CNNLSTM	0.03	2.2	17.00	20.0	70.83	74.9	27.42	48.3
Combining Multiple Methods and Algorithms	1-class AutoEncoder	0.99	3.2	**21.00**	22.2	8.57	68.9	12.17	48.5
All Features	MTHC	0.03	2.5	7.00	25.6	43.75	66.2	12.07	36.9
Feature Voting Ranking	MTMS (no age-aware)	0.03	3.9	18.00	**43.2**	72.00	78.2	28.80	55.6
Age-aware Mechanism	MTL	**0.01**	**1.8**	13.00	36.7	**81.25**	80.0	22.41	50.1
Age-aware Mechanism	MTMS	0.03	2.0	19.33	41.6	76.67	**87.2**	**30.87**	**56.3**

deepen our analysis of age-related failure indicators beyond POH and explore ways to reduce label dependency while enhancing model transferability to unseen datasets.

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